Practical 11: Data Modelling and Analytics with Pivot Table in Excel

Data Model is used for building a model where data from various sources can be combined by creating relationships among the data sources. A Data Model integrates the tables, enabling extensive analysis using PivotTables, Power Pivot, and Power View.

A **Data Model** is created automatically when you import two or more tables simultaneously from a database. The existing database relationships between those tables is used to create the Data Model in Excel.

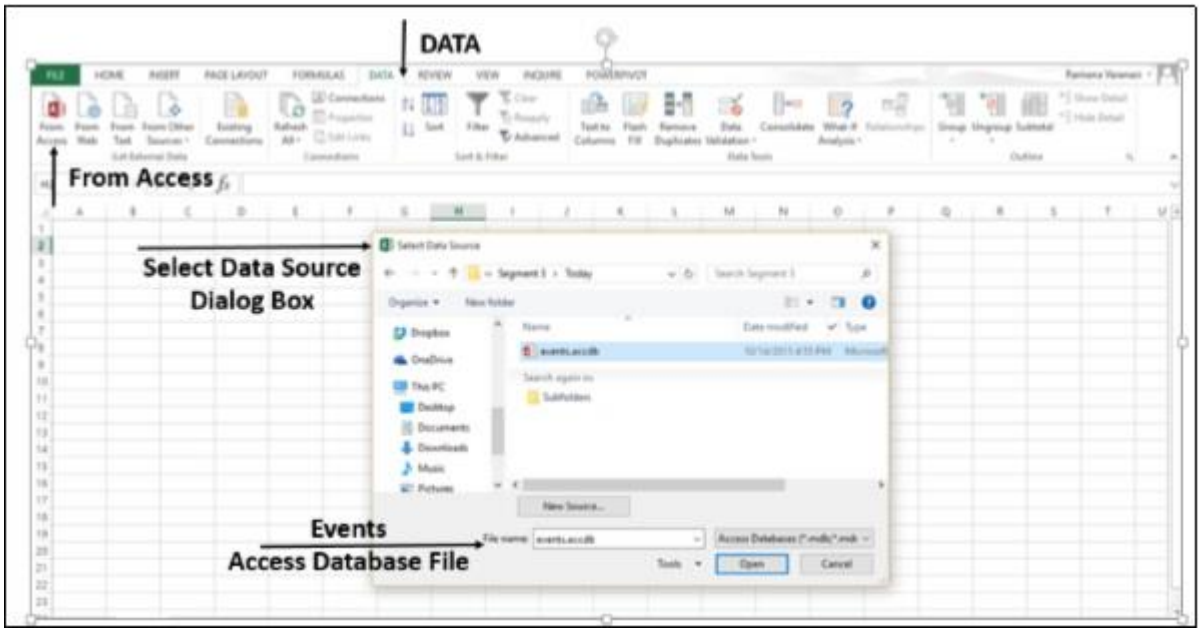
Step 1 − Open a new blank Workbook in Excel.

Step 2 − Click on the **DATA** tab.

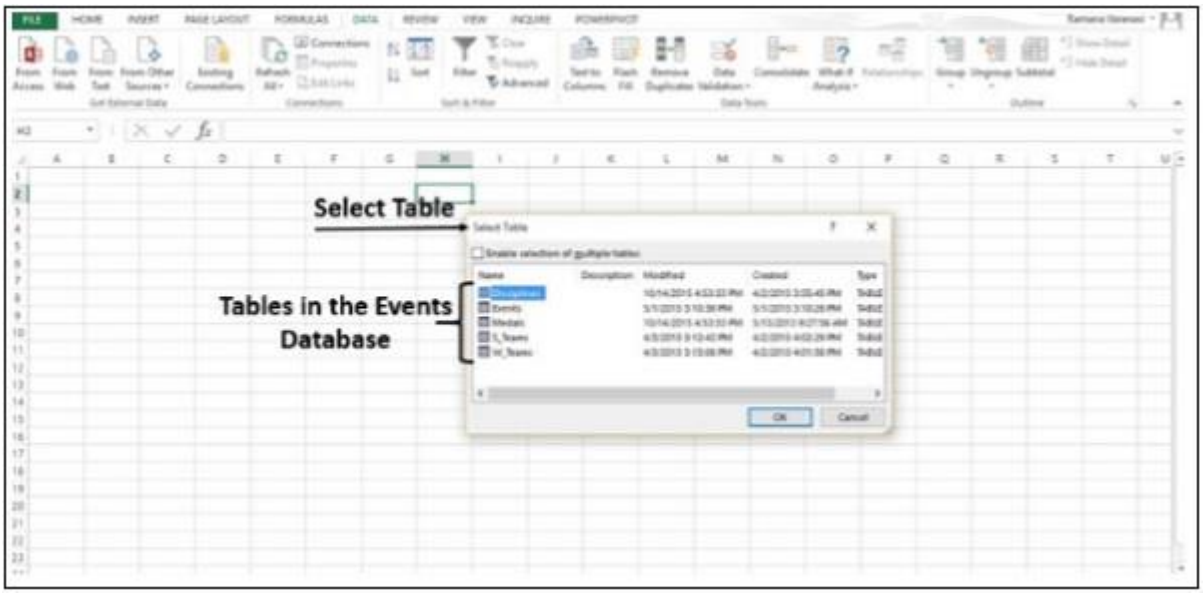
Step 3 − In the **Get External Data** group, click on the option **From Access**. The **Select Data Source** dialog box opens.

Step 4 − Select **Events.accdb**, Events Access Database file.

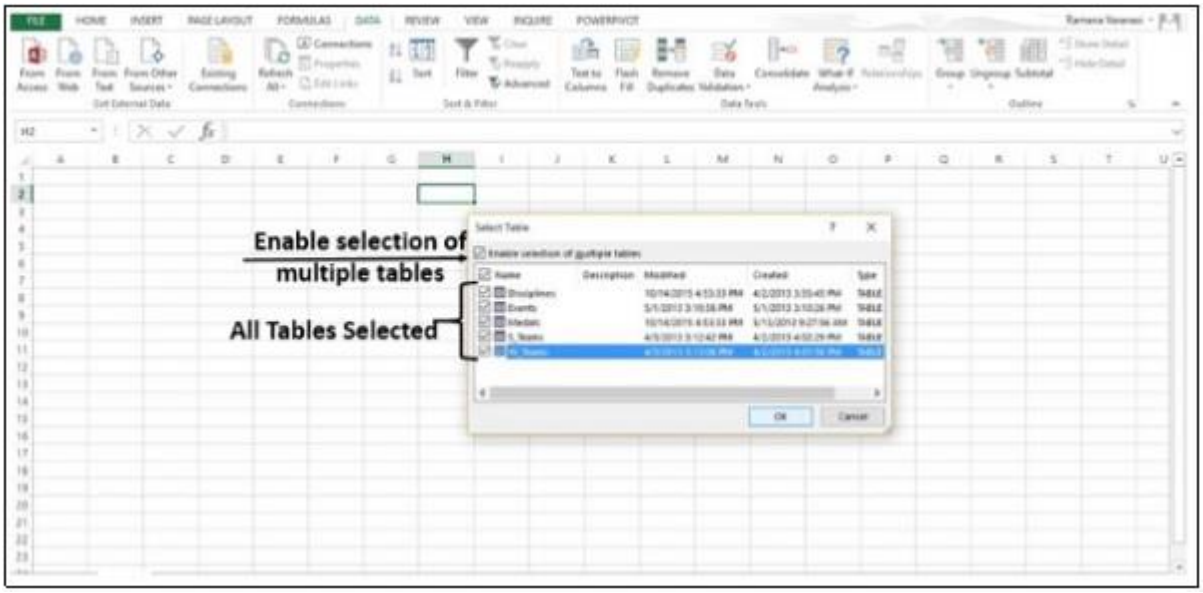
(url:/ https://fcschools.instructure.com/courses/373/files/10607)



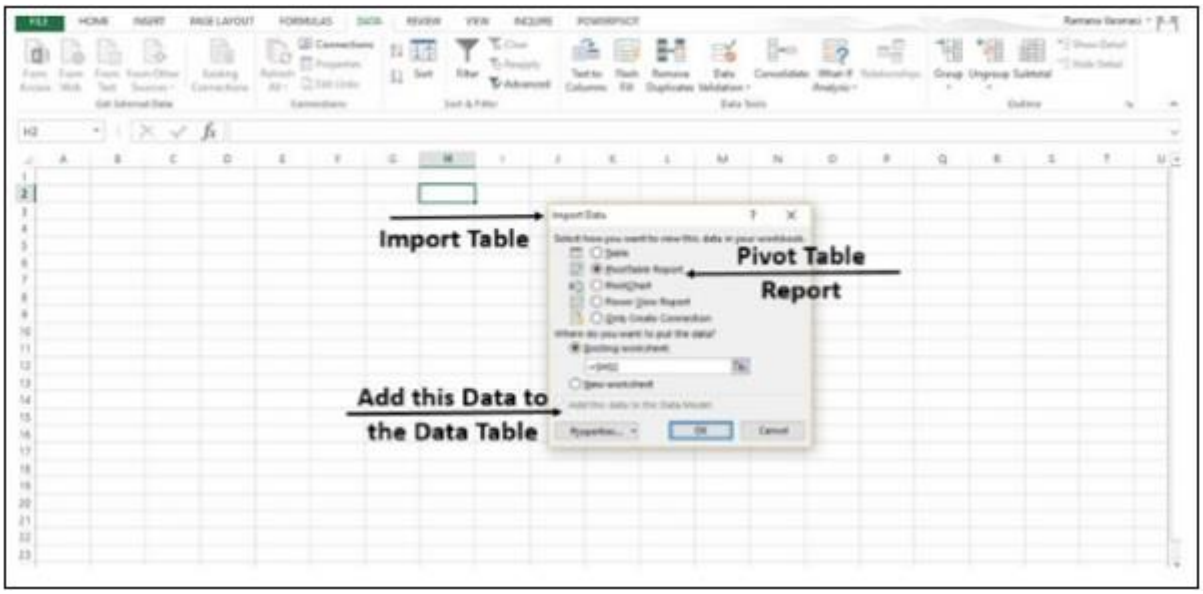
Step 5 − The **Select Table** window, displaying all the **tables** found in the database, appears.



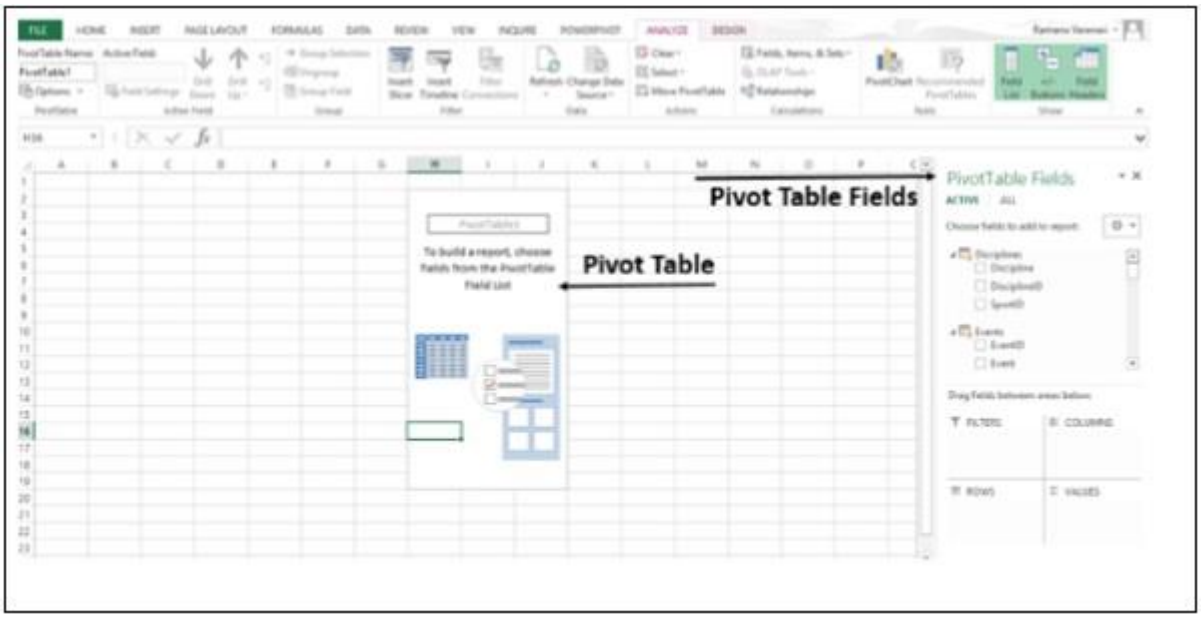
Step 6 − Tables in a database are similar to the tables in Excel. Check the **'Enable selection of multiple tables'** box, and select all the tables. Then click **OK**.



Step 7 − The **Import Data** window appears. Select the **PivotTable Report** option. This option imports the tables into Excel and prepares a PivotTable for analyzing the imported tables. Notice that the checkbox at the bottom of the window - **'Add this data to the Data Model'** is selected and disabled.



Step 8 − The data is imported, and a **PivotTable** is created using the imported tables.



Explore Data Using PivotTable

Step 1 − You know how to add fields to PivotTable and drag fields across areas. Even if you are not sure of the final report that you want, you can play with the data and choose the best-suited report.

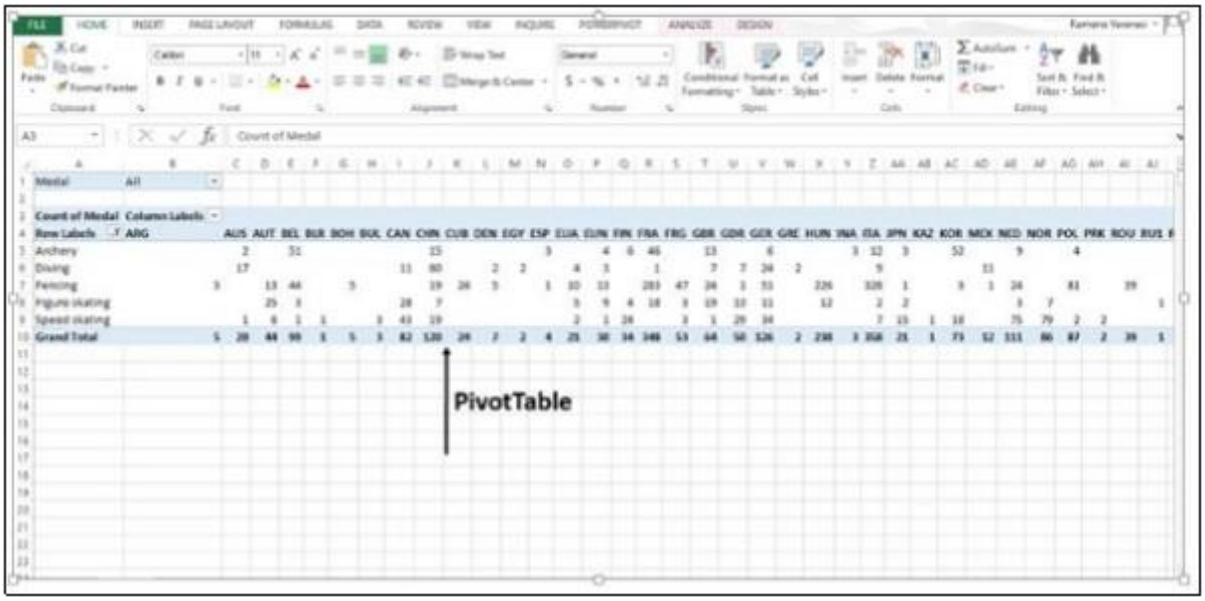
In **PivotTable Fields**, click on the arrow beside the table **- Medals** to expand it to show the fields in that table. Drag the **NOC_CountryRegion** field in the **Medals** table to the **COLUMNS** area.

Step 2 − Drag **Discipline** from the **Disciplines** table to the **ROWS** area.

Step 3 − Filter **Discipline** to display only five sports: Archery, Diving, Fencing, Figure Skating, and Speed Skating. This can be done either in **PivotTable Fields** area, or from the **Row Labels** filter in the PivotTable itself.

Step 4 − In **PivotTable Fields**, from the **Medals** table, drag Medal to the **VALUES** area.

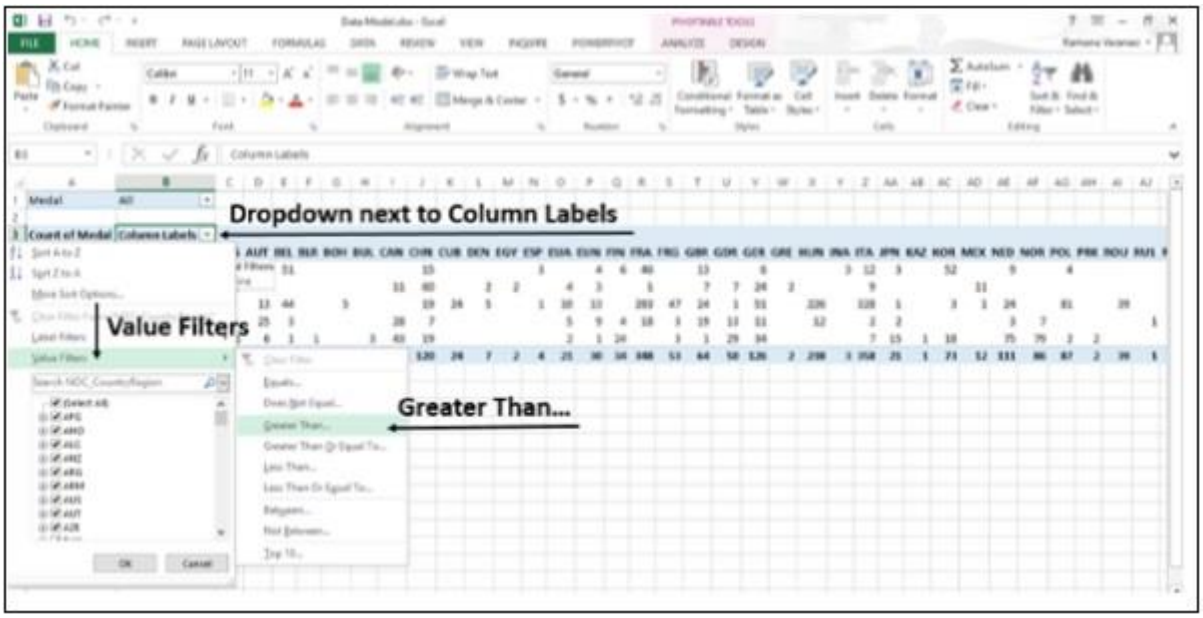
Step 5 − From the **Medals** table, select **Medal** again and drag it into the **FILTERS** area.



Step 6 − Click the dropdown list button to the right of the **Column** labels.

Step 7 − Select **Value Filters** and then select **Greater Than**…

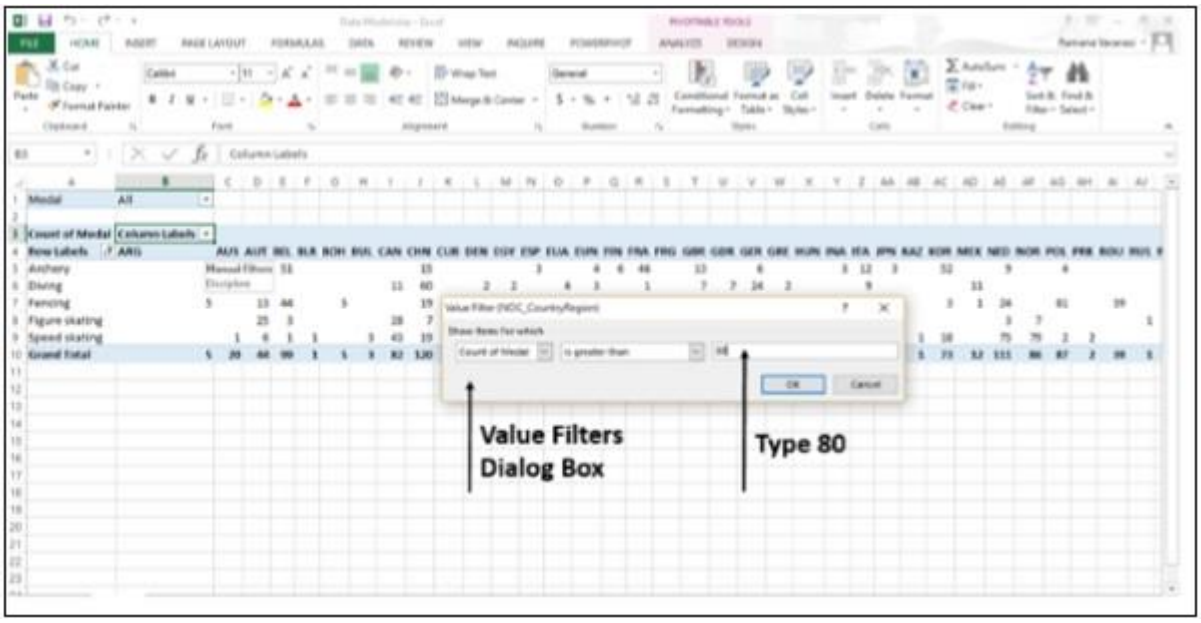
Step 8 − Click **OK**.



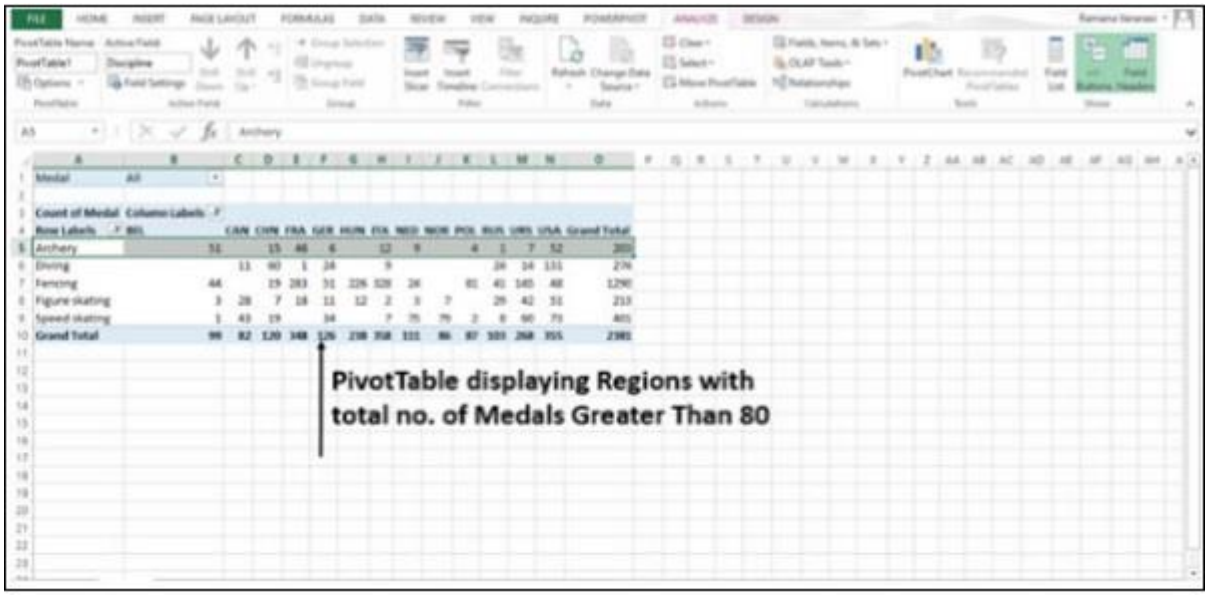
The Value **Filters dialog** box for the count of Medals **is greater than** appears.

Step 9 – Type **80** in the **Right Field**.

Step 10 – Click **OK**.



The PivotTable displays only those regions, which has more than total 80 medals.

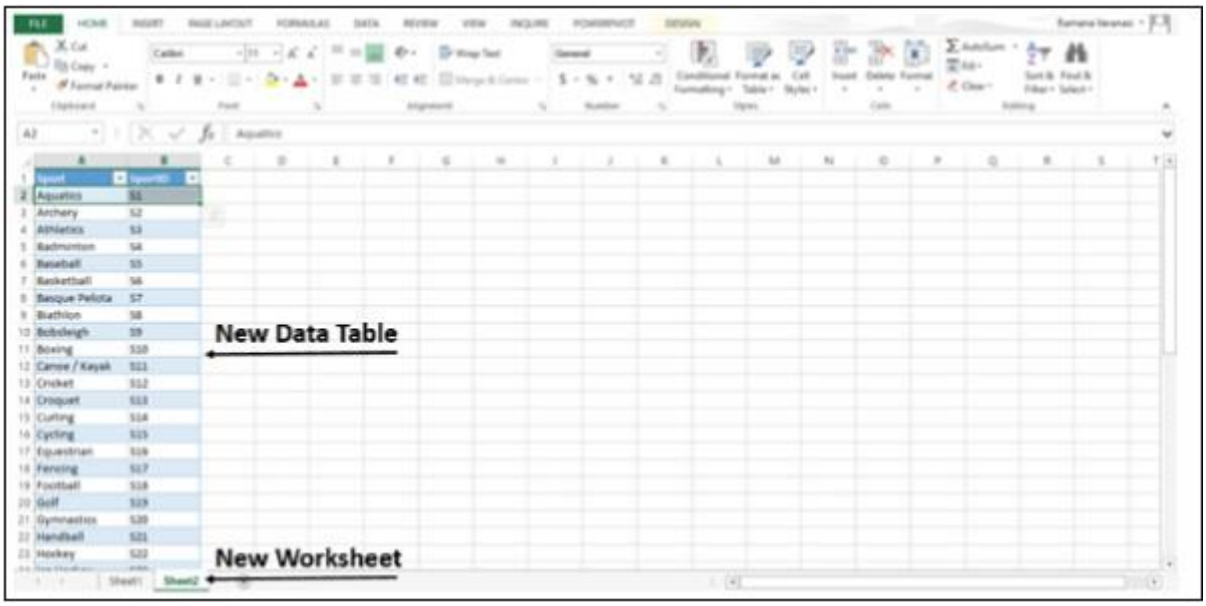


Create Relationship between Tables

Relationships let you analyze your collections of the data in Excel, and create interesting and aesthetic reports from the data you import.

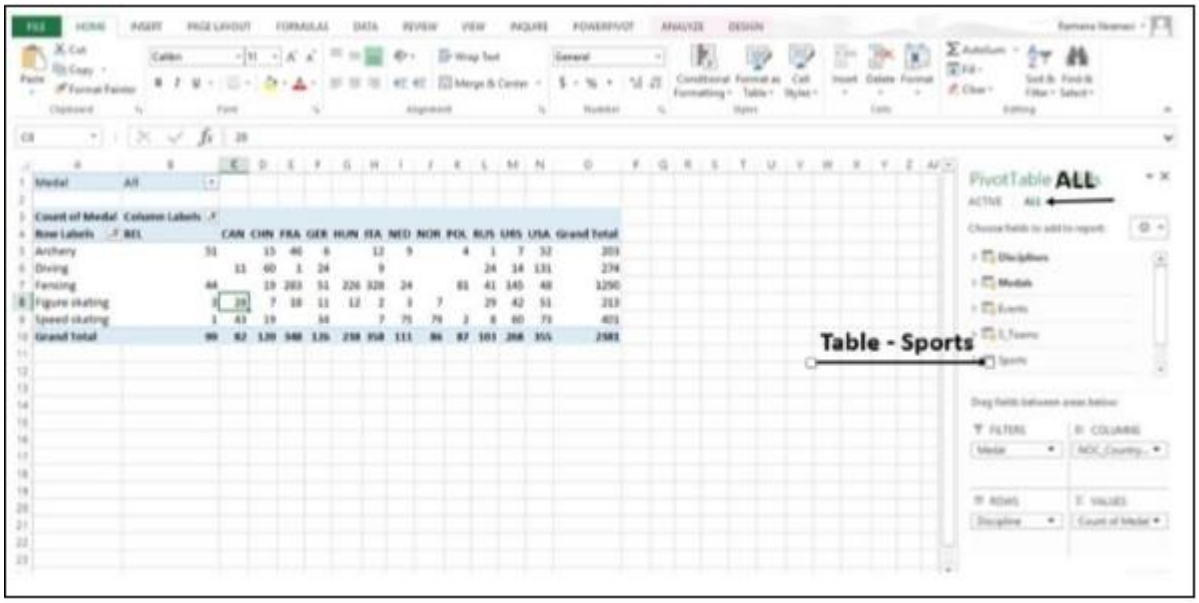
Step 1 – **Insert** a new Worksheet.

Step 2 – Create a new table with new data. Name the new table as **Sports**.

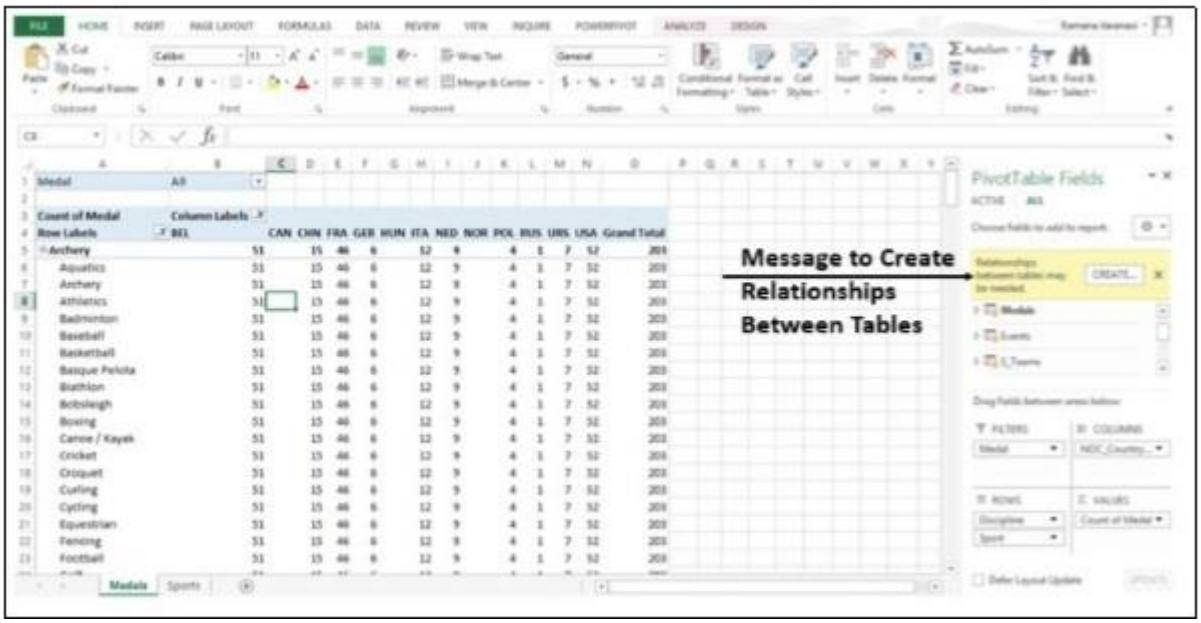


Step 3 − Now you can create relationship between this new table and the other tables that already exist in the **Data Model** in Excel. Rename the Sheet1 as **Medals** and Sheet2 as **Sports**.

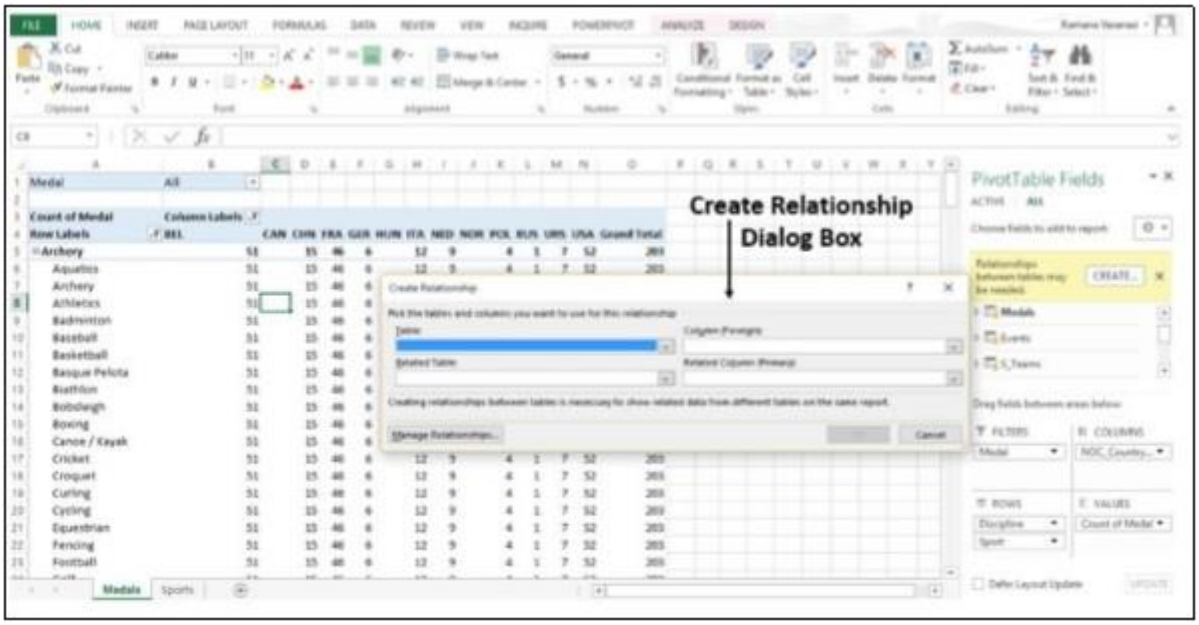
On the **Medals sheet**, in the **PivotTable Fields List**, click **All**. A complete list of available tables will be displayed. The newly added table - **Sports** will also be displayed.



Step 4 − Click on **Sports**. In the expanded list of fields, select **Sports**. Excel messages you to create a relationship between tables.



Step 5 – Click on **CREATE**. The **Create Relationship** dialog box opens.



Step 6 − To create the relationship, one of the tables must have a column of unique, non-repeated, values. In the **Disciplines** table, **SportID** column has such values. The table **Sports** that we have created also has the **SportID** column. In **Table**, select **Disciplines**.

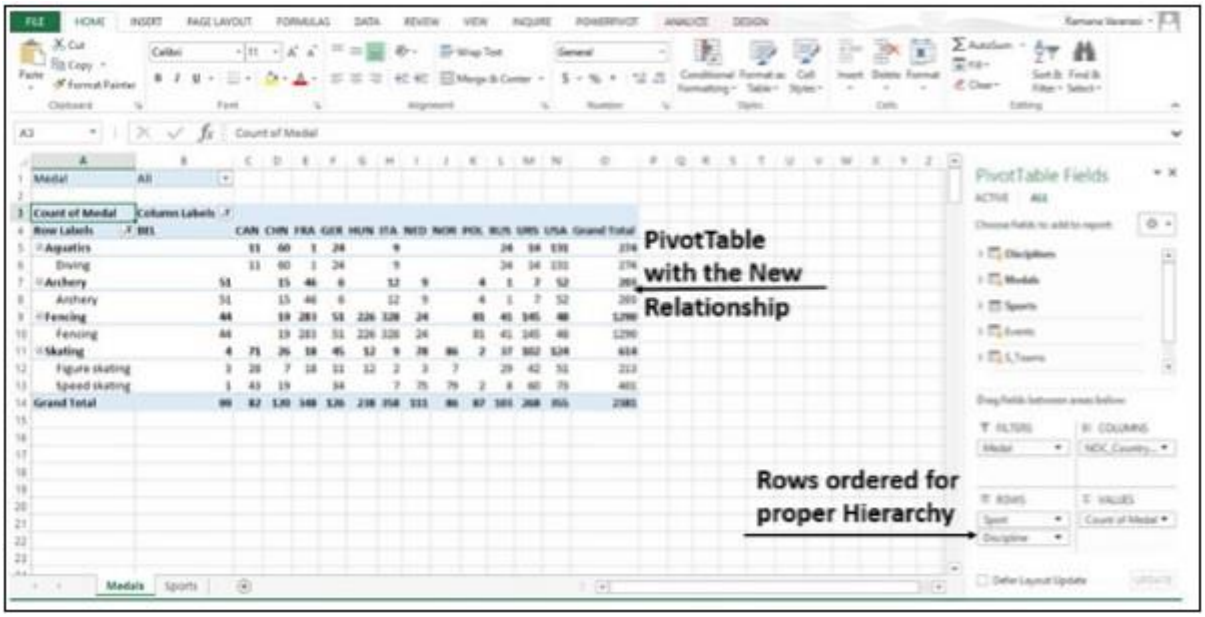
Step 7 − In **Column (Foreign)**, select SportID.

Step 8 − In **Related Table**, select **Sports**.

Step 9 − In **Related Column (Primary)**, SportID gets selected automatically. Click **OK**.

Step 10 − The **PivotTable** is modified to reflect the addition of the new **Data Field** Sport. Adjust the order of the fields in the Rows area to maintain the **Hierarchy**. In this case, **Sport**

should be first and **Discipline** should be the next, as **Discipline** will be nested in Sport as a sub-category.



Practical 12: Data Analysis and Visualization using Advanced Excel

Power View is a feature of Microsoft Excel 2013 that enables **interactive** data exploration, visualization, and presentation encouraging intuitive ad-hoc reporting.

Create a Power View Sheet

Make sure **Power View** add-in is enabled in Excel 2013.

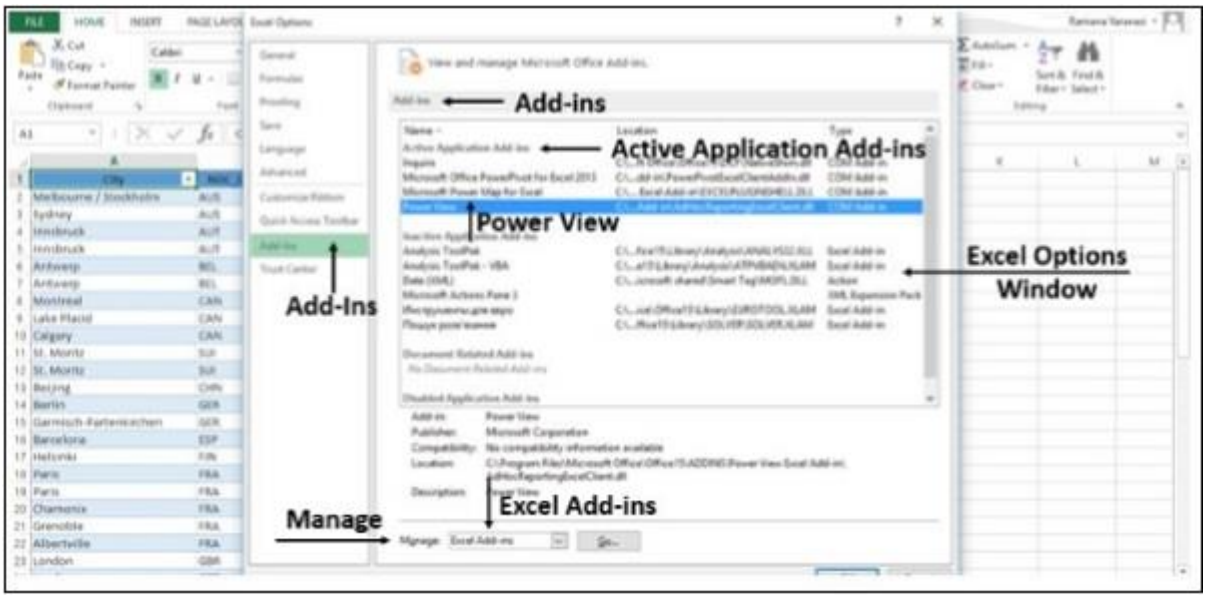
Step 1 − Click on the **File** menu and then Click on **Options**.

The **Excel Options** window appears.

Step 2 − Click on **Add-Ins**.

Step 3 − In the **Manage** box, click the drop-down arrow and select **Excel Add-ins**.

Step 4 − All the available **Add-ins** will be displayed. If **Power View** Add-in is enabled, it appears in Active Application Add-ins.



If it does not appear, follow these steps −

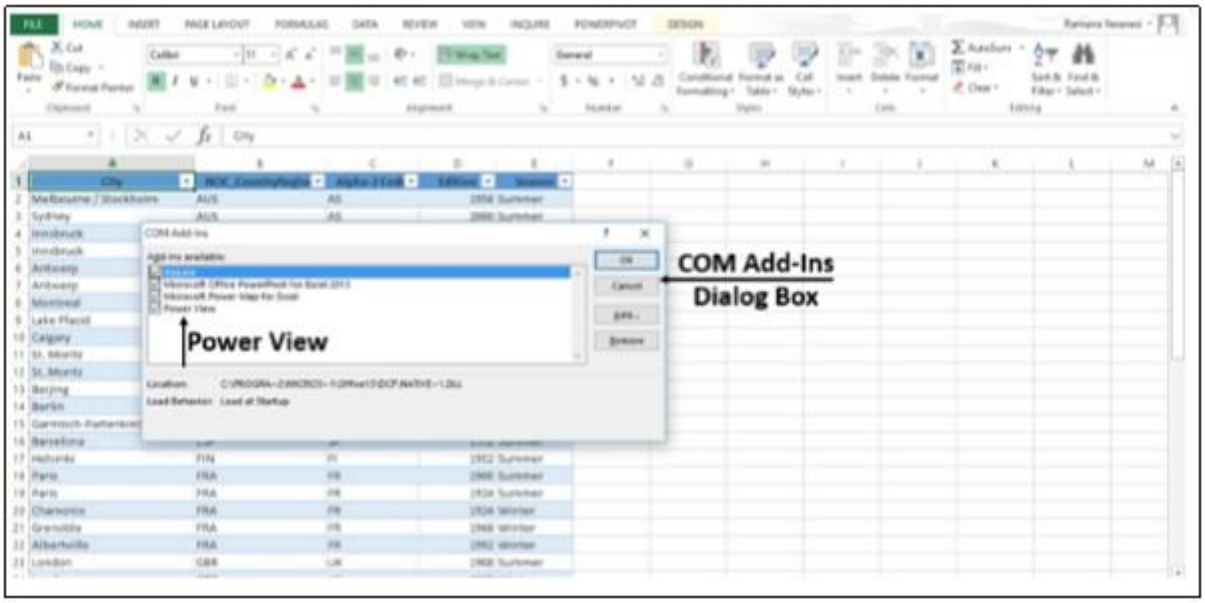
Step 1 − In the **Excel Options** Window, Click on **Add-Ins**.

Step 2 − In the **Manage** box, click the drop-down arrow and select **COM Add-ins**

Step 3 − Click on the **Go** button. A **COM Add-Ins** Dialog Box appears.

Step 4 − Check the **Power View** Check Box.

Step 5 − Click **OK**.

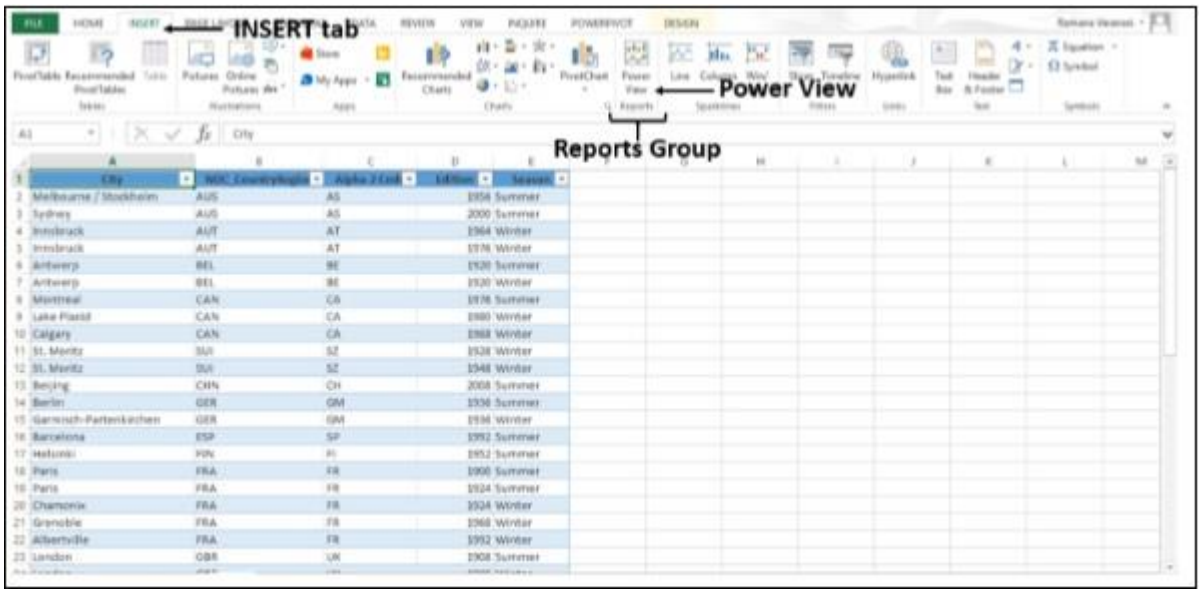


Now, you are ready to create the **Power View sheet**.

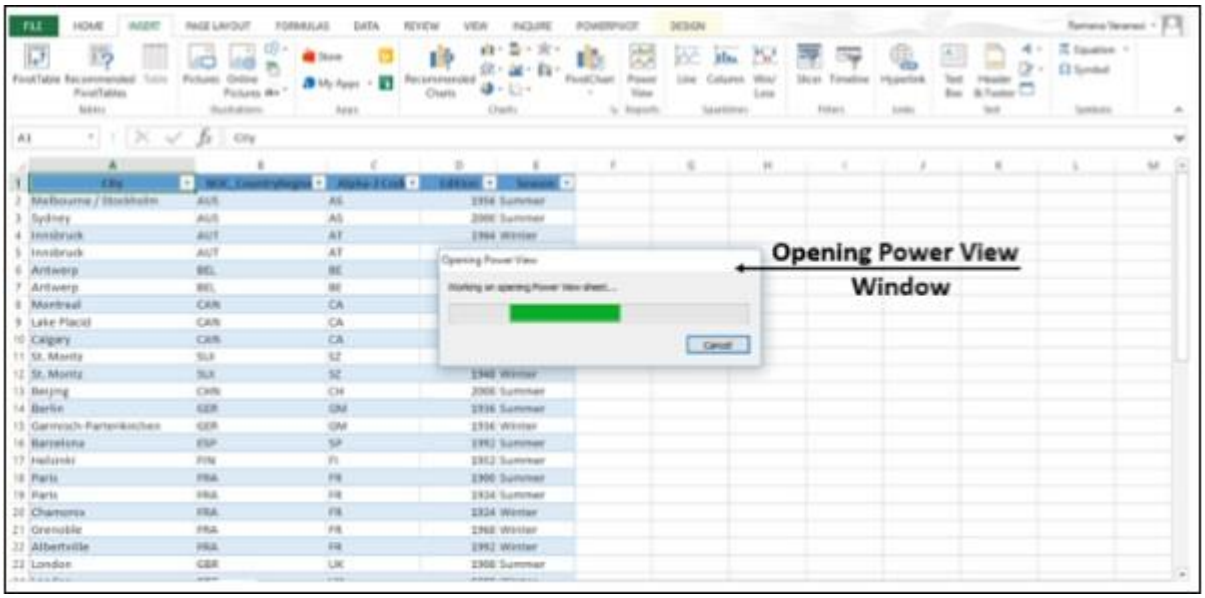
Step 1 − Click on the **Data Table**.

Step 2 − Click on **Insert** tab.

Step 3 − Click on **Power View** in **Reports** Group.

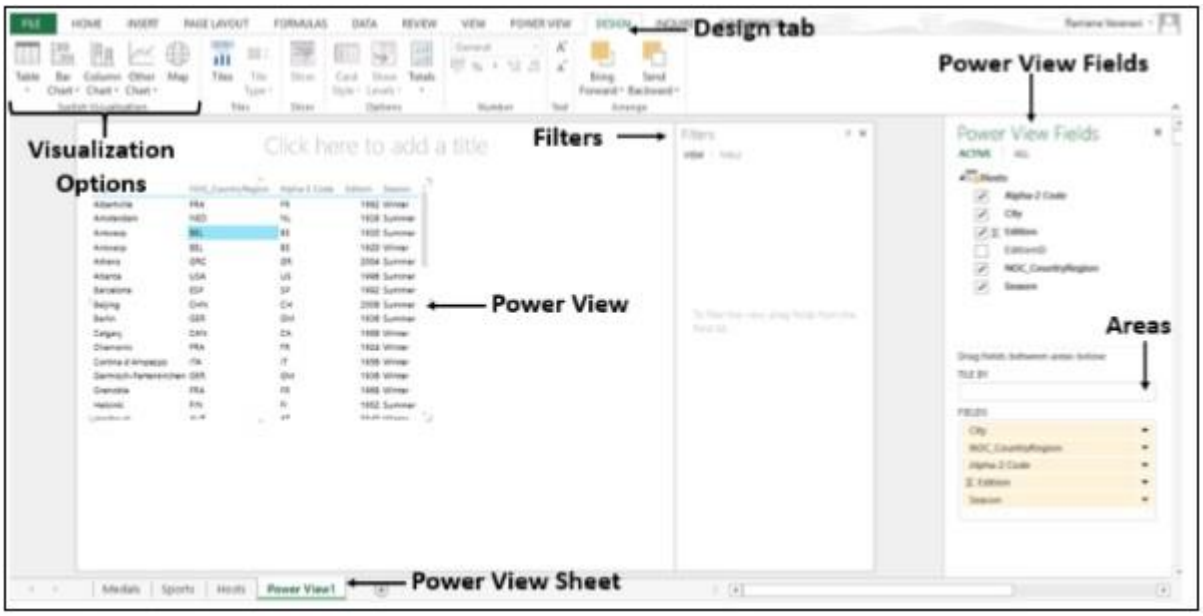


An **Opening Power View** window opens, showing the progress of Working on opening Power View sheet.



The **Power View sheet** is created for you and added to your Workbook with the **Power View**. On the Right-side of the **Power View**, you find the **Power View Fields**. Under the **Power View Fields** you will find **Areas**.

In the Ribbon, if you click on **Design tab**, you will find various **Visualization** options.



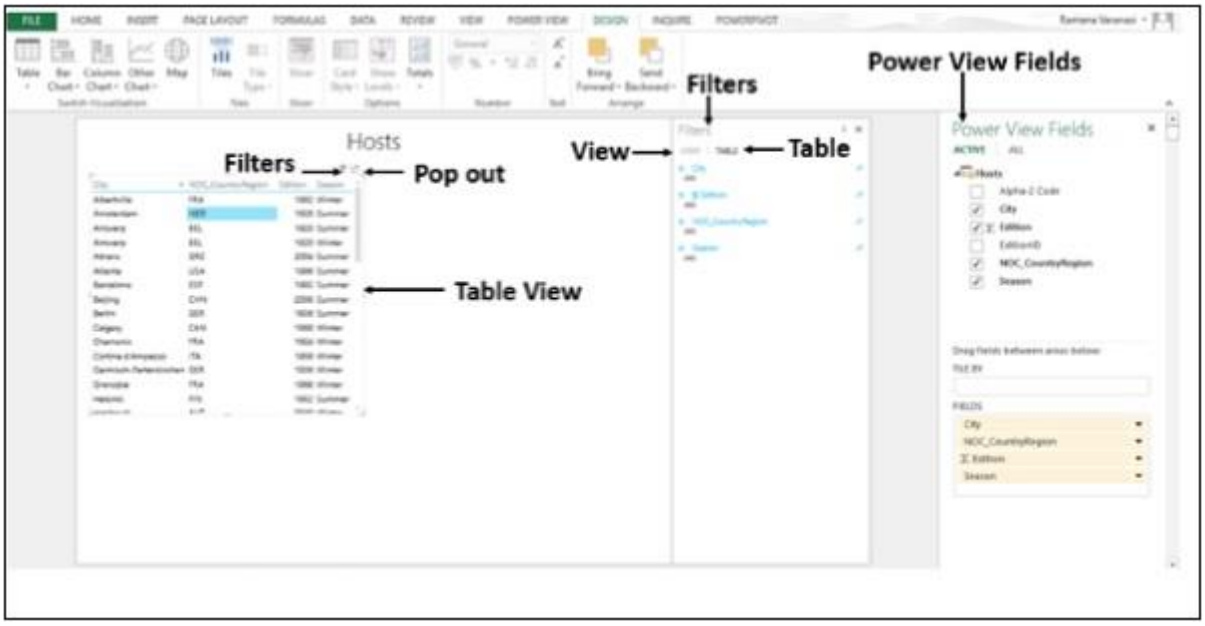
Create Charts and other Visualizations

For every visualization you want to create, you start on a Power View sheet by creating a table, which you then easily convert to other visualizations, to find one that best illustrates your Data.

Step 1 − Under the **Power View Fields**, select the fields you want to visualize.

Step 2 − By default, the **Table** View will be displayed. As you move across the Table, on the top-right corner, you find two symbols – Filters and Pop out.

Step 3 − Click on the **Filters** symbol. The filters will be displayed on the right side. **Filters** has two tabs. View tab to filter all visualizations in this **View** and **Table** tab to filter the specific values in this table only.



Visualization – Matrix

A **Matrix** is made up of rows and columns like a **Table**. However, a Matrix has the following capabilities that a Table does not have −

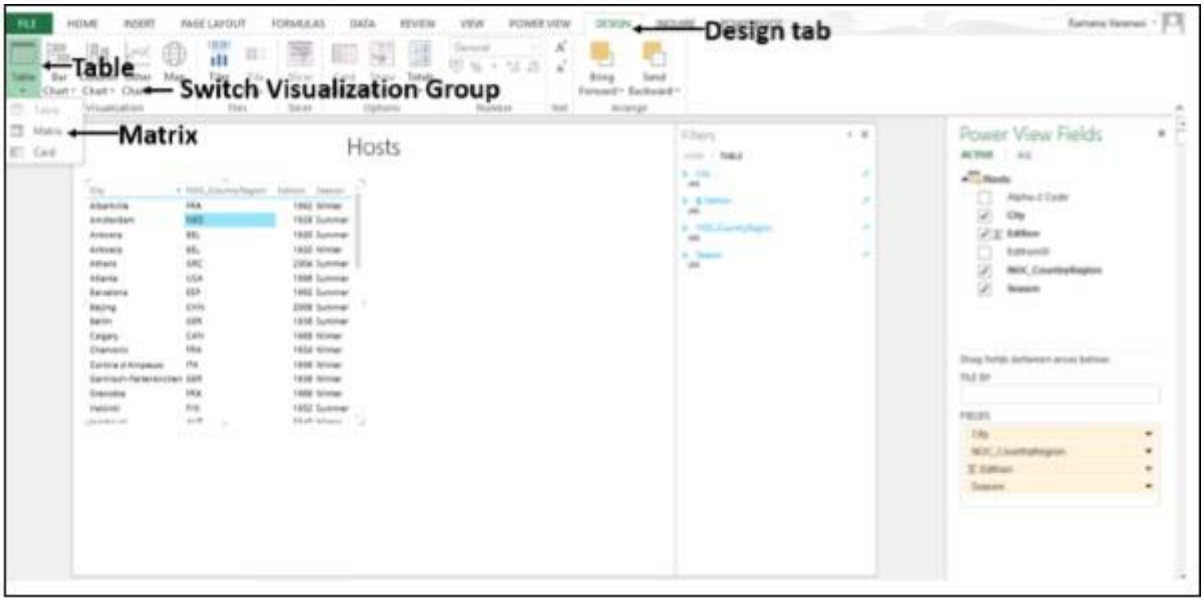
- Display data without repeating values.
- Display totals and subtotals by row and column.
- With a hierarchy, you can drill up/drill down.

Collapse and Expand the Display

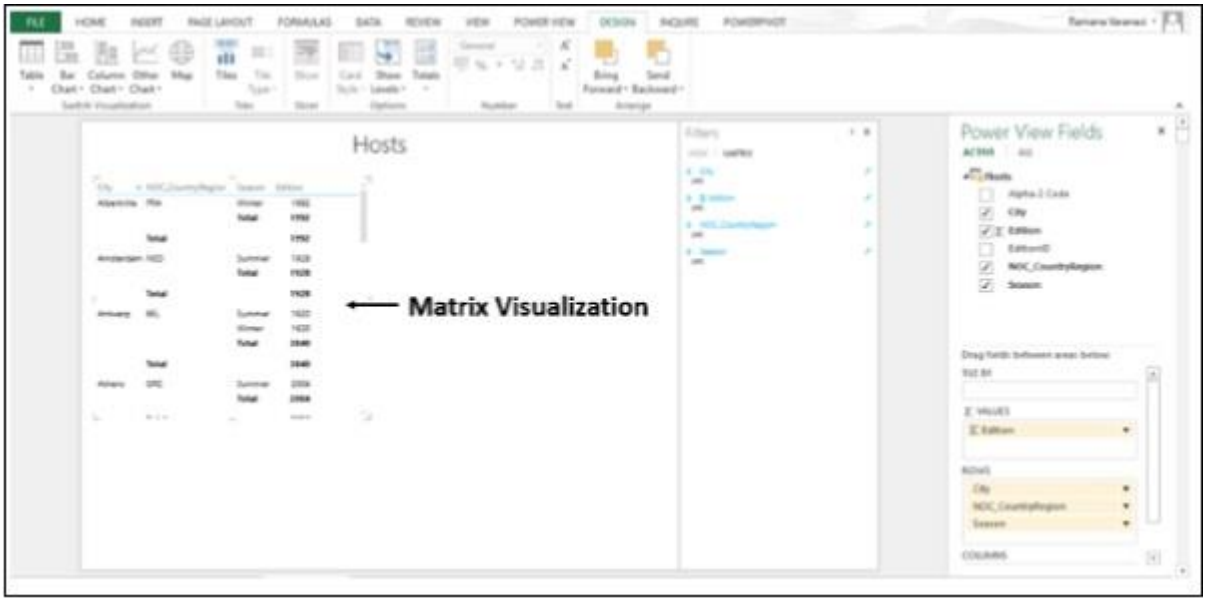
Step 1 − Click on the **DESIGN** tab.

Step 2 − Click on **Table** in the **Switch Visualization** Group.

Step 3 − Click on **Matrix**.



The **Matrix Visualization** appears.



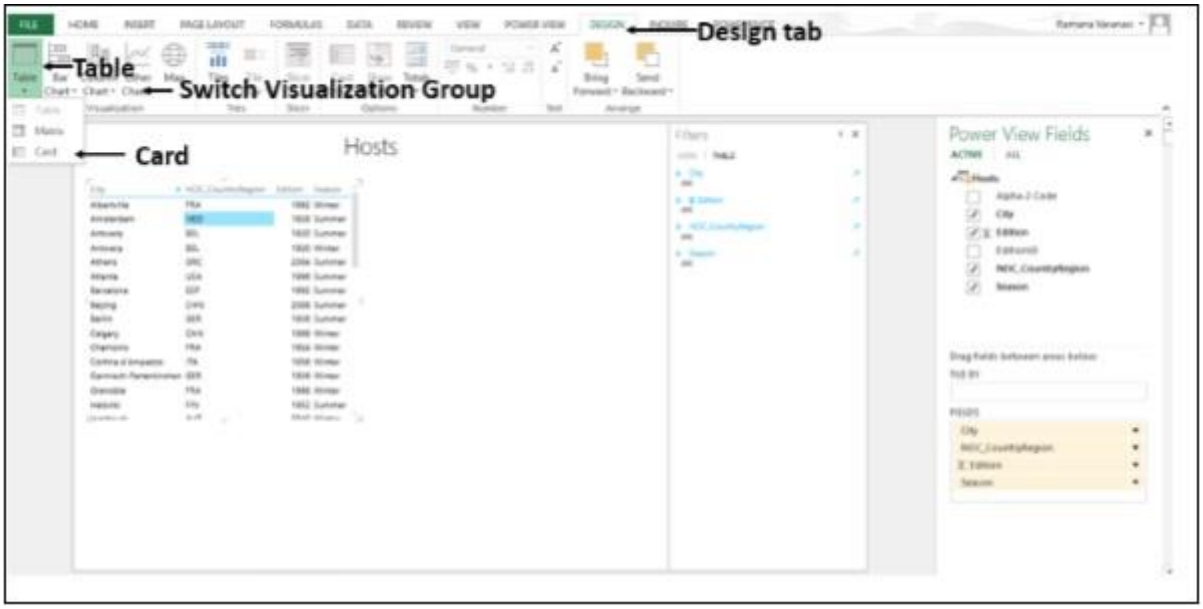
Visualization – Card

You can convert a **Table** to a series of **Cards** that display the data from **each row** in the table laid out in a **Card format**, like an **index Card**.

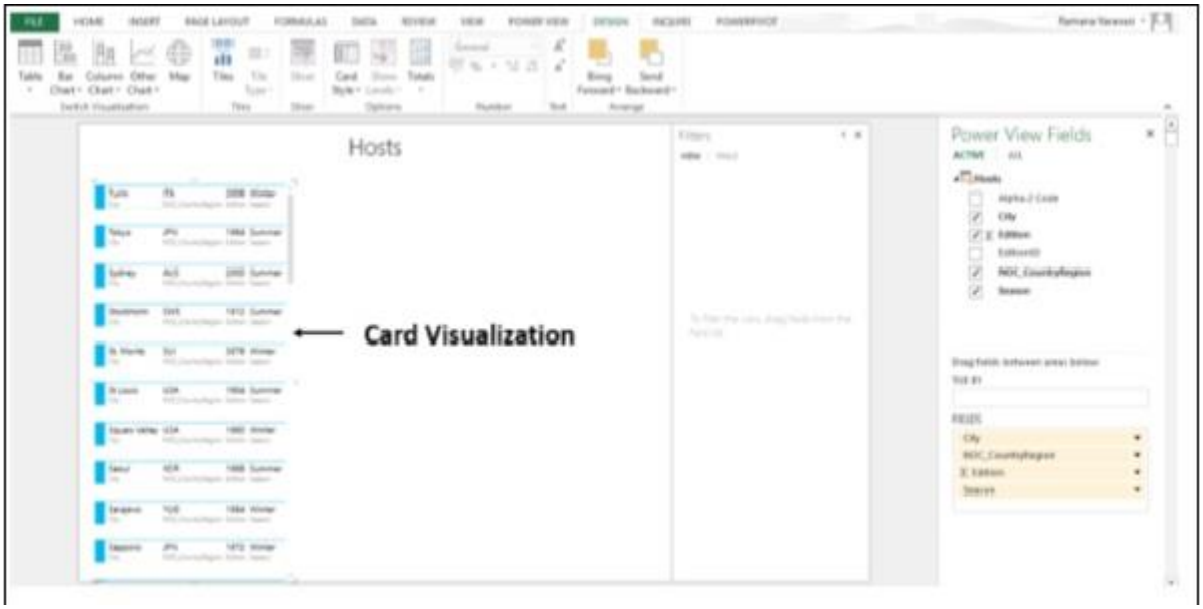
Step 1 − Click on the **DESIGN** tab.

Step 2 − Click on **Table** in the **Switch Visualization** Group.

Step 3 − Click on **Card**.



The **Card Visualization** appears.



Visualization – Charts

In **Power View**, you have a number of Chart options: Pie, Column, Bar, Line, Scatter, and Bubble. You can use several design options in a chart such as showing and hiding labels, legends, and titles.

Charts are interactive. If you click on a Value in one Chart −

- the **Value** in that chart is highlighted.
- All the Tables, Matrices, and Tiles in the report are filtered to that Value.
- That **Value** in all the other Charts in the report is highlighted.

The charts are interactive in a presentation setting also.

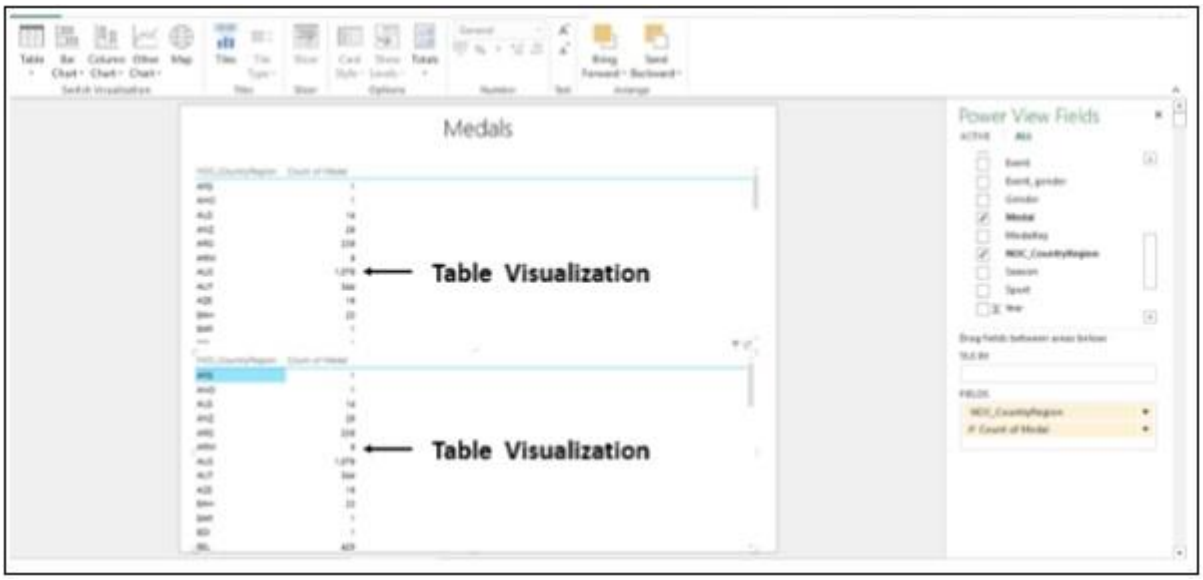
Step 1 − Create a **Table Visualization** from **Medals** data.

You can use Line, Bar and Column Charts for comparing data points in one or more data series. In these Charts, the x-axis displays one field and the y-axis displays another, making it easy to see the relationship between the two values for all the items in the Chart.

Line Charts distribute category data evenly along a horizontal (category) axis, and all numerical value data along a vertical (value) axis.

Step 2 − Create a Table Visualization for two Columns, **NOC_CountryRegion** and **Count of Medal**.

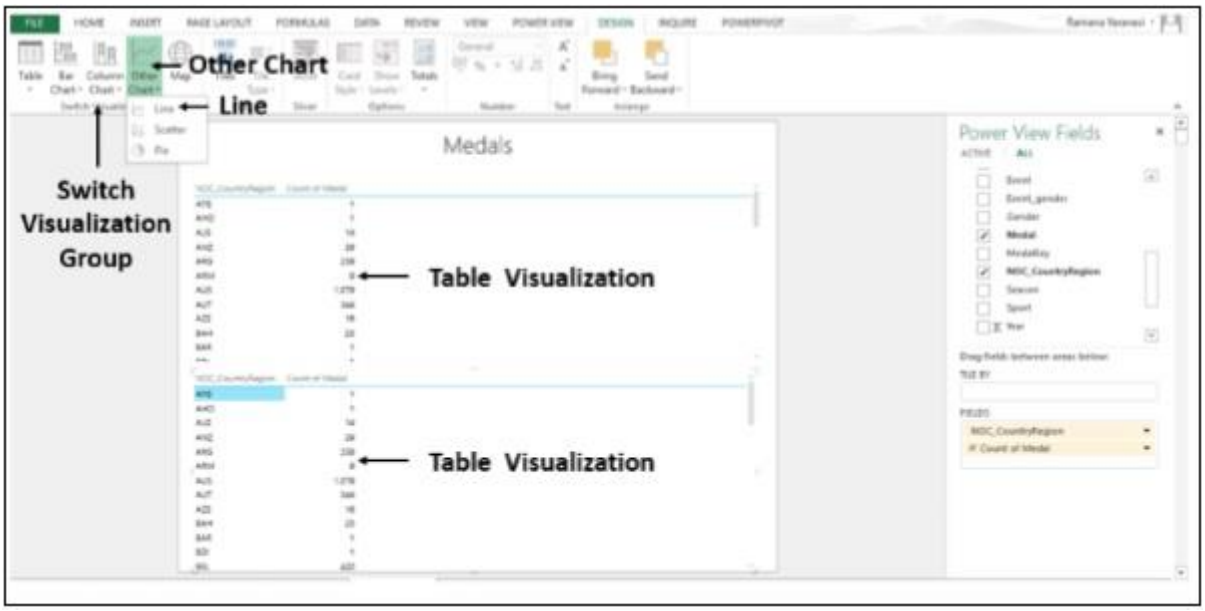
Step 3 − Create the same Table Visualization below.



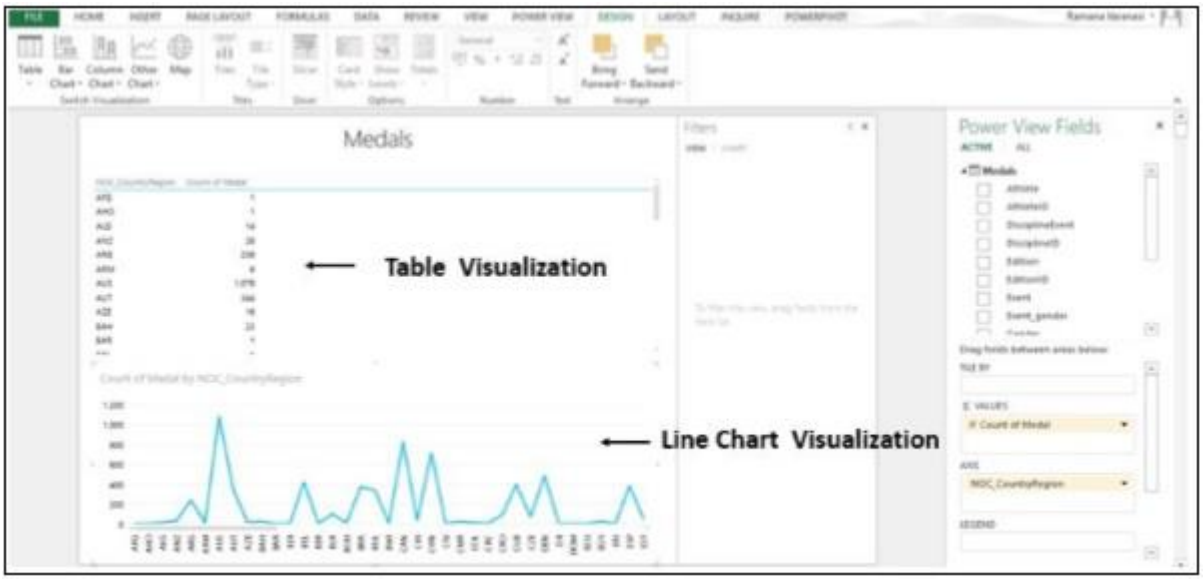
Step 4 − Click on the **Table Visualization** below.

Step 5 − Click on **Other Chart** in the **Switch Visualization** group.

Step 6 − Click on **Line**.



The **Table Visualization** converts into **Line Chart Visualization**.

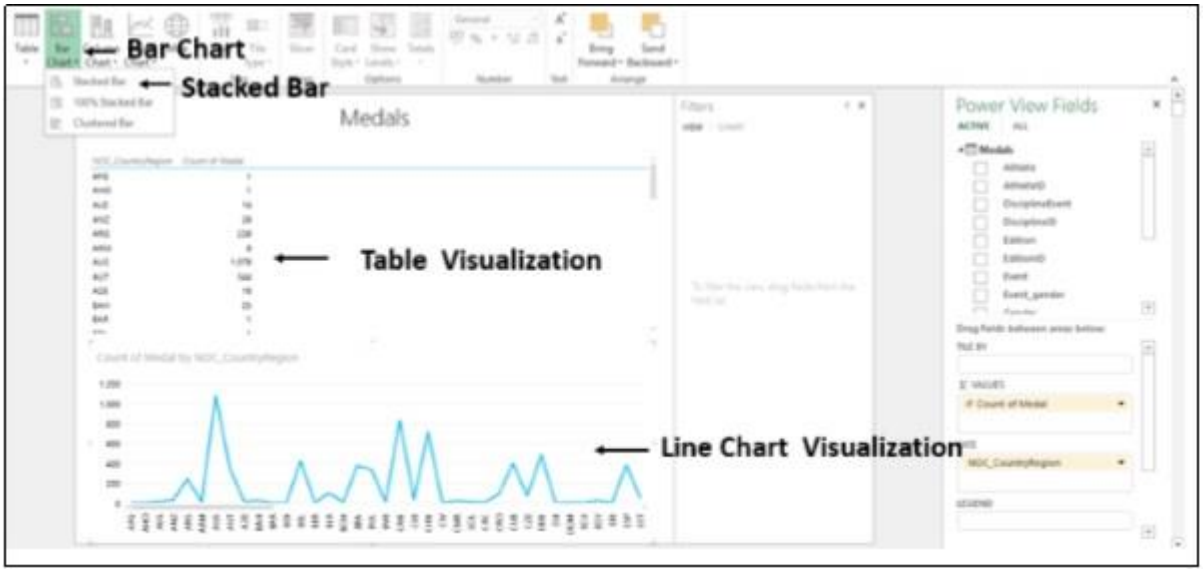


In a **Bar Chart**, categories are organized along the vertical axis and values along the horizontal axis. In **Power View**, there are three subtypes of the **Bar Chart: Stacked, 100% stacked**, and **Clustered**.

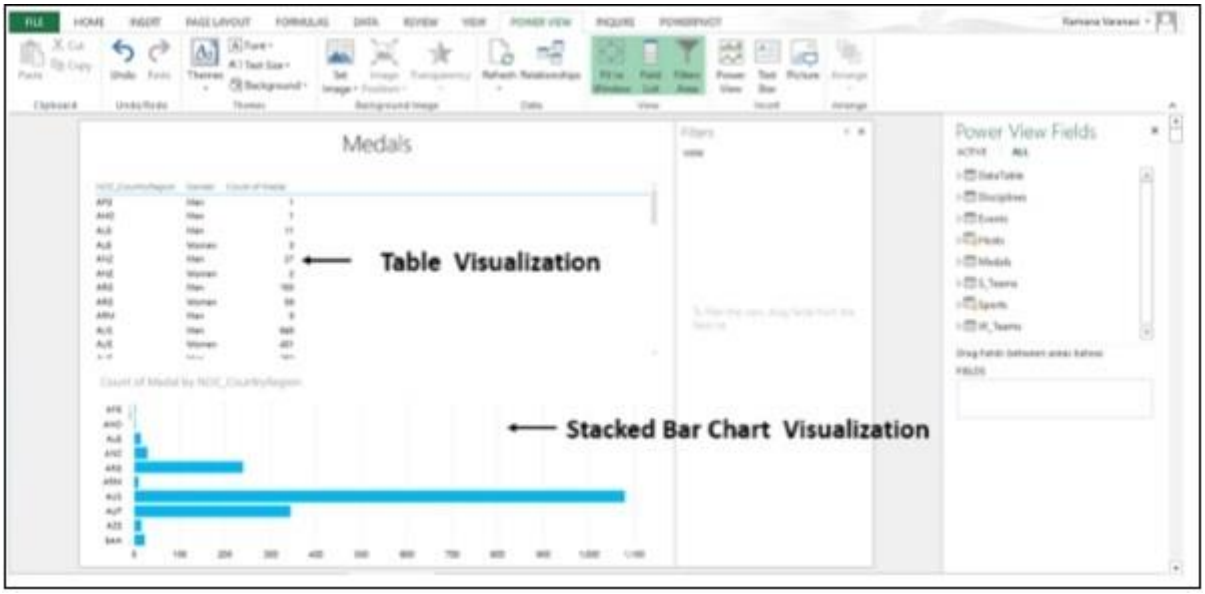
Step 7 − Click on the **Line Chart Visualization**.

Step 8 − Click on **Bar Chart** in the **Switch Visualization** Group.

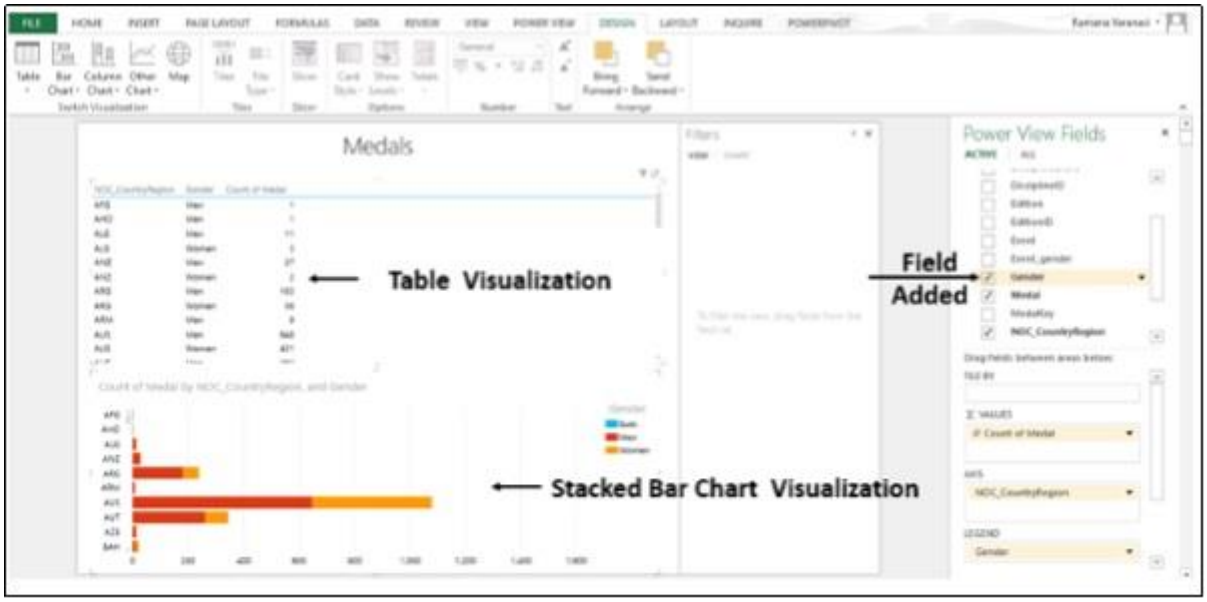
Step 9 − Click on the **Stacked Bar** option.



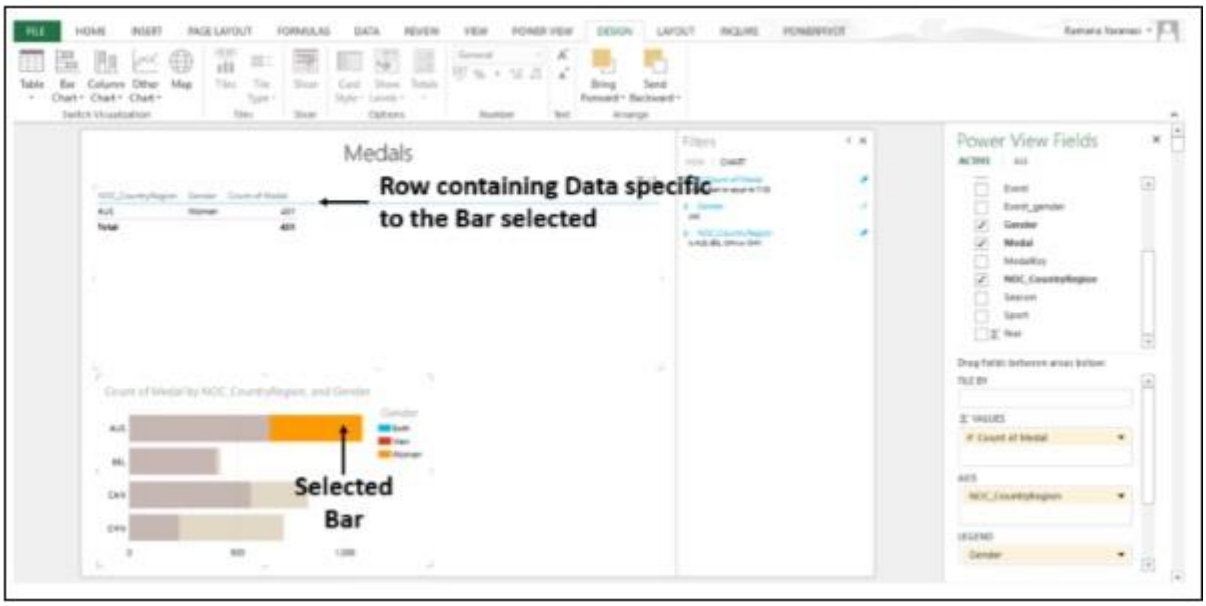
The **Line Chart Visualization** converts into **Stacked Bar Chart Visualization**.



Step 10 – In the **Power View Fields**, in the **Medals** Table, select the **Field Gender** also.



Step 11 – Click on one of the bars. That portion of the bar is highlighted. Only the row containing the Data specific to the selected bar is displayed in the table above.



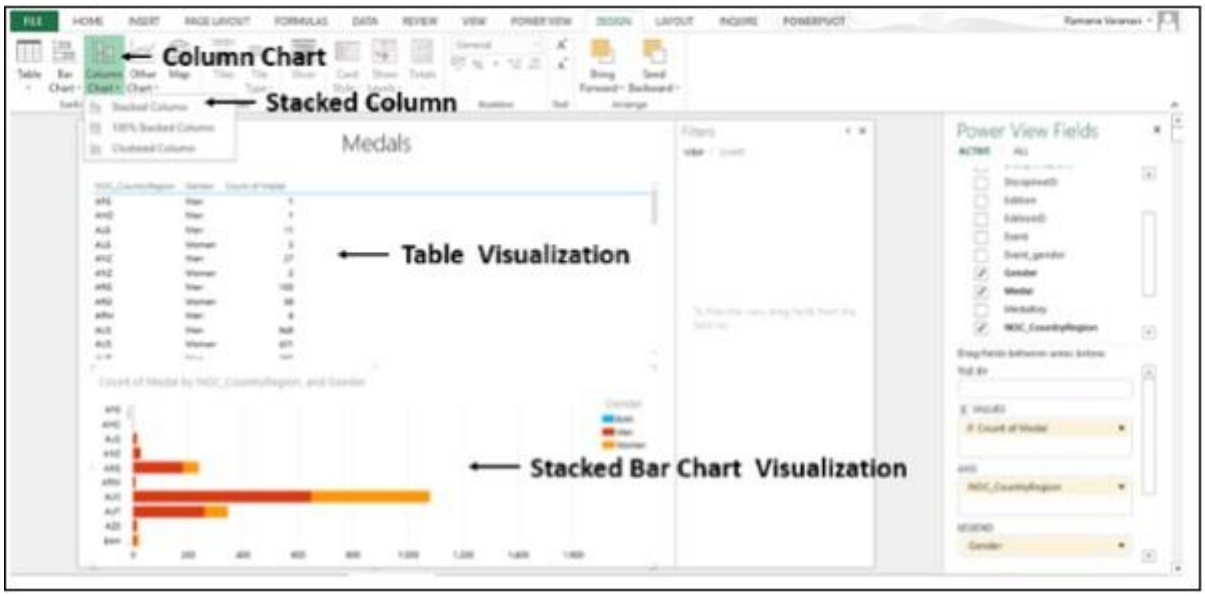
You can use the column charts for showing data changes over a period of time or for illustrating comparison among different items. In a Column Chart, the categories are along the horizontal axis and values are along the vertical axis.

In Power View, there are three **Column Chart** subtypes: **Stacked, 100% stacked**, and **Clustered**.

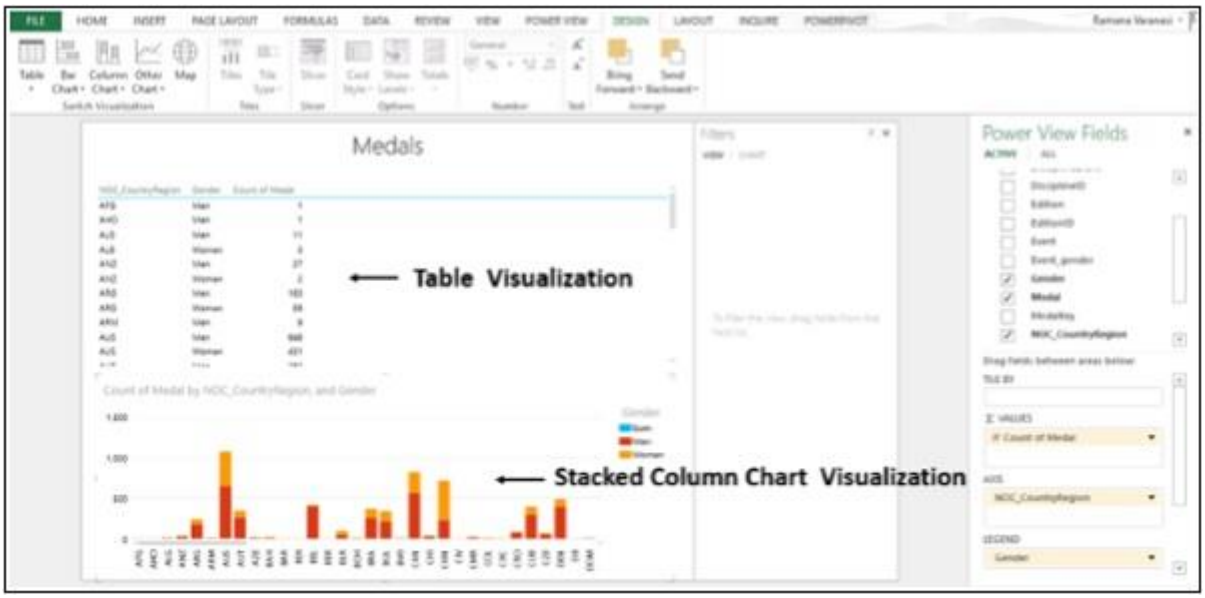
Step 12 − Click on the **Stacked Bar Chart Visualization**.

Step 13 − Click on Column Chart in the **Switch Visualization** group.

Step 14 − Click on **Stacked Column**.

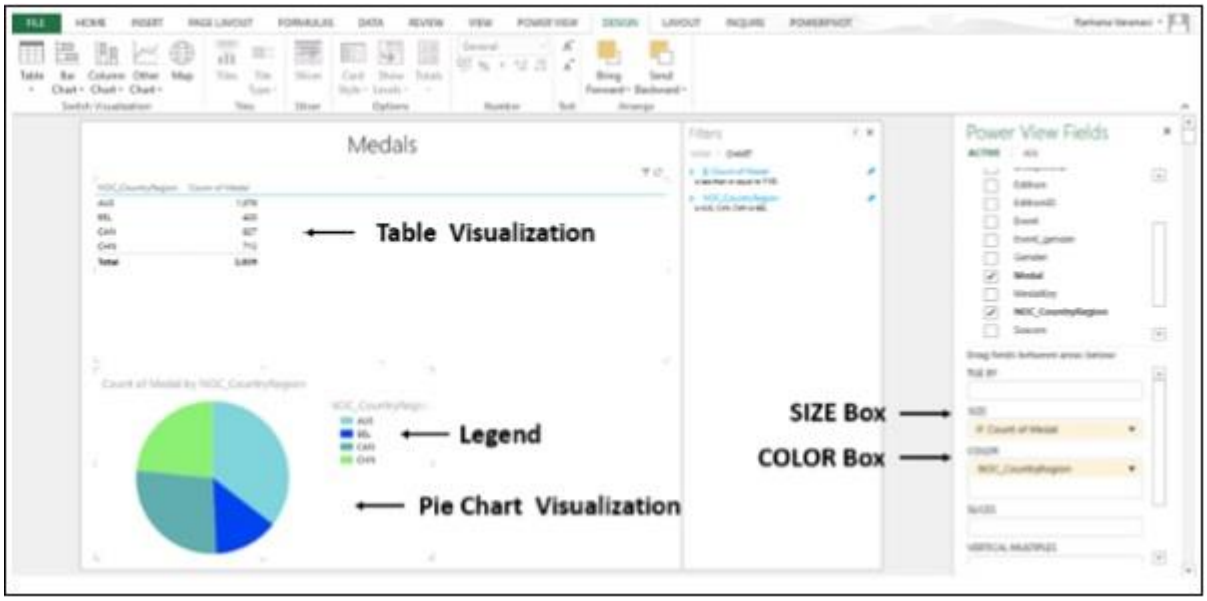


The **Stacked Bar Chart** Visualization converts into **Stacked Column Chart** Visualization.



You can have simple **Pie Chart Visualizations** in Power View.

Step 1 − Click on the **Table Visualization** as shown below.

Step 2 − Click on **Other Chart** in the **Switch Visualization** group.

Step 3 − Click on **Pie** as shown in the image given below.